

College of Science and Technology Royal University of Bhutan

Computing Technologies Department

Bachelor of Engineering in Information Technology Autumn Semester 2026

CTE412 – Integrated System Submitted by:

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Std. no: 02230147

Task 1

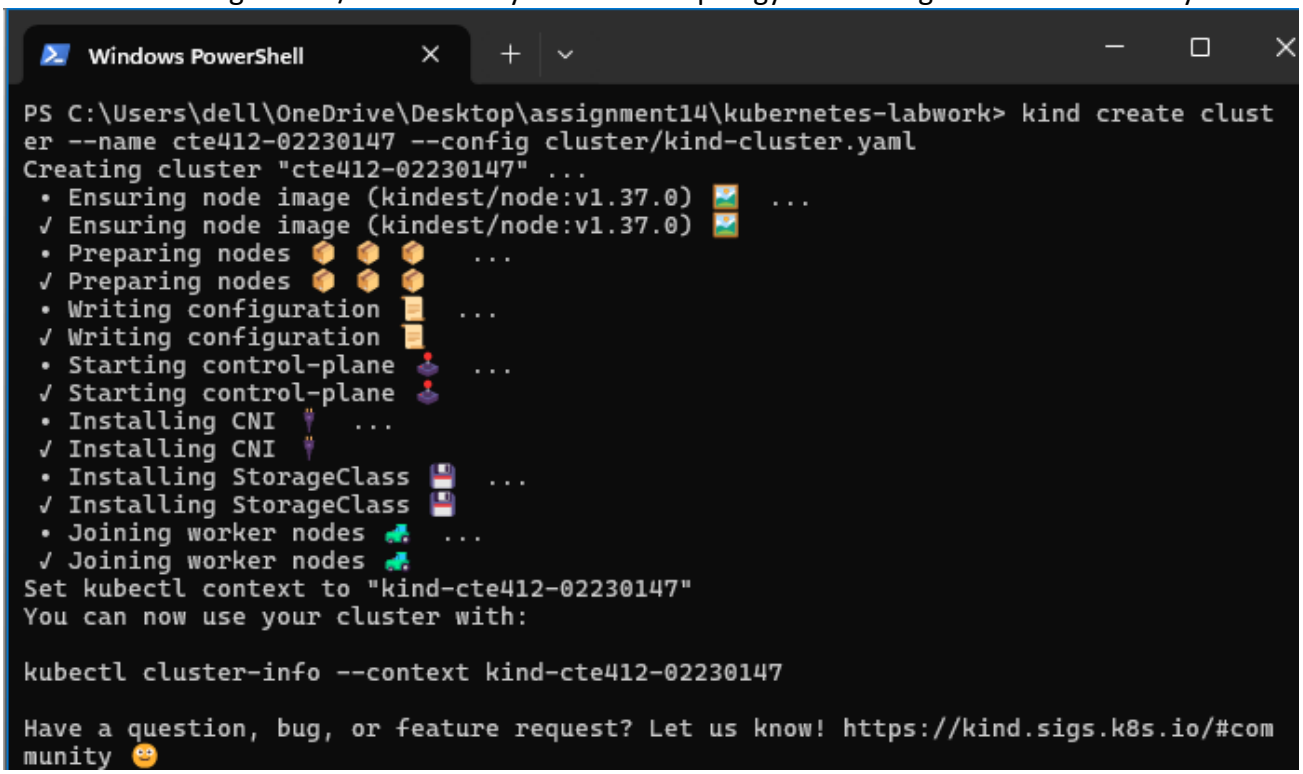
1. Create a kind cluster named cte412-<sid> with one control-plane node and two worker nodes. Host port 30090, bound to 127.0.0.1 only, must reach node port 30090 on the cluster.

Kind-cluster.yaml file

```yaml
kind: Cluster
apiVersion: kind.x-k8s.io/v1alpha4
name: cte412-02230129
nodes:
  - role: control-plane
    extraPortMappings:
      - containerPort: 30090 # NodePort inside the kind node
        hostPort: 30090 # published on the Windows host
        listenAddress: "127.0.0.1" # loopback only, per brief
        protocol: TCP
  - role: worker # notice-board pods
  - role: worker # pod spread + multi-pod /version responses
```

Commands:

- kind create cluster: Tells the KinD CLI to spin up a new local Kubernetes cluster
- --name cte412-02230147: sets the name of the cluster to cte412-02230147.
- --config cluster/kind-cluster.yaml: Reads topology and settings from kind-cluster.yaml



2. Create the namespace nb-02230147 with the labels app.kubernetes.io/part-of=campus-noticeboard and cte412/practical="06", and an annotation recording your full name.

```yaml
apiVersion: v1
kind: Namespace
metadata:
  name: nb-02230147
  labels:
    app.kubernetes.io/part-of: campus-notice-board # requ
    cte412/practical: "06" # required by brief (string)
  annotations:
    student.fullname: "Sherab Dorji" # required by brief
```

3. In that namespace, create a ResourceQuota with these hard limits: requests.cpu 1, requests.memory 1Gi, limits.cpu 2, limits.memory 2Gi, pods 15, count/services 4.

```yaml
# Purpose: hard quota sized for 3 replicas + monitor +
# Apply: kubectl apply -f manifests/base/01-quota.yaml
apiVersion: v1
kind: ResourceQuota
metadata:
  name: nb-quota
  namespace: nb-02230147
spec:
  hard:
    requests.cpu: "1" # 6 app pods (600m) + monitor (10
    requests.memory: 1Gi
    limits.cpu: "2"
    limits.memory: 2Gi
    pods: "15" # 6 app + monitor + surge + terminating
    count/services: "4" # notice-board, notice-board-ex
```

4. Create a LimitRange supplying default requests and limits for containers. Choose the values yourself and justify each one in a comment against the quota above.

```yaml
apiVersion: v1
kind: LimitRange
metadata:
  name: nb-defaults
  namespace: nb-02230147
spec:
  limits:
  - type: Container
    default:
      cpu: 200m # <= limits.cpu ceiling; 6 pods × 200m = 1.2 < 2
      memory: 256Mi # 6 pods × 256Mi = 1.5Gi < 2Gi ceiling
    defaultRequest:
      cpu: 100m # 6 pods × 100m = 600m < requests.cpu=1
      memory: 128Mi # 6 pods × 128Mi = 768Mi < 1Gi
```

5. Make nb-<sid> the default namespace of your kubectl context.

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifest
s/base/00-namespace.yaml
namespace/nb-02230147 created
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifest
s/base/01-quota.yaml
resourcequota/nb-quota created
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifest
s/base/02-limitrange.yaml
limitrange/nb-defaults created
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl config set-contex
t --current --namespace=nb-02230147
Context "kind-cte412-02230147" modified.
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get nodes -o wide

NAME                              STATUS   ROLES           AGE     VERSION   INTERNAL-IP   EXT
ERNAL-IP   OS-IMAGE                         KERNEL-VERSION                          CONTAIN
ER-RUNTIME
cte412-02230147-control-plane     Ready    control-plane   6m38s   v1.37.0   172.25.0.3    <no
ne>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd64)   contain
erd://2.3.4
cte412-02230147-worker            Ready    <none>          6m10s   v1.37.0   172.25.0.2    <no
ne>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd64)   contain
erd://2.3.4
cte412-02230147-worker2           Ready    <none>          6m10s   v1.37.0   172.25.0.4    <no
ne>        Debian GNU/Linux 13 (trixie)   6.6.87.2-microsoft-standard-WSL2 (amd64)   contain
erd://2.3.4
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl describe quota nb-quota -n nb-02230147
Name:           nb-quota
Namespace:      nb-02230147
Resource        Used  Hard
--------        ----  ----
count/services  0     4
limits.cpu      0     2
limits.memory   0     2Gi
pods            0     15
requests.cpu    0     1
requests.memory 0     1Gi
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl config view --minify | Select-String "n
amespace"

    namespace: nb-02230147
```

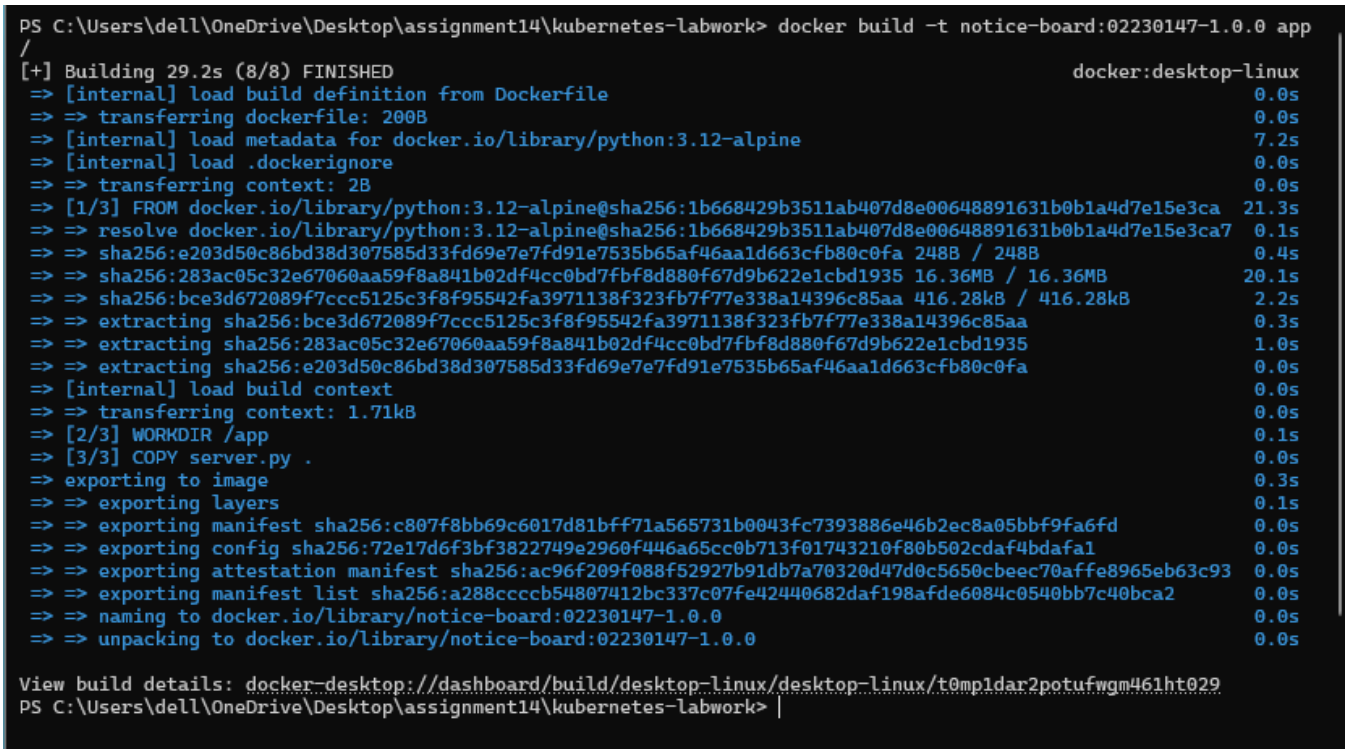
Evidence:

```
                          STATUS   ROLES           AGE   VERSION   INTERNAL-IP   EXTERNAL-IP
02230147-control-plane    Ready    control-plane   2m5s  v1.37.0   172.24.0.4    <none>
02230147-worker           Ready    <none>          103s  v1.37.0   172.24.0.6    <none>
02230147-worker2          Ready    <none>          103s  v1.37.0   172.24.0.7    <none>
```

```
1  Name:            nb-quota
2  Namespace:       nb-02230147
3  Resource         Used  Hard
4  --------         ----  ----
5  count/services   0     4
6  limits.cpu       0     2
7  limits.memory    0     2Gi
8  pods             0     15
9  requests.cpu     0     1
10 requests.memory  0     1Gi
11
```

Task 2

1. Every version listen on port 8080, runs as a non-root user, and exposes three endpoints. GET /notices returns at least three notices as JSON. GET /version returns JSON containing version, student (your <sid>) and pod (the container hostname). GET /healthz returns HTTP 200 when healthy.
2. docker tag noticeboard-app:v1.0.0 notice-board: 02230147-1.0.0
3. Tag notice-board:<sid>-1.1.0 — adds a category field to every notice and a pinned Boolean to at least one. When the environment variable FEATURE_PINNED equals true, pinned notices are listed first and the response body includes "pinnedFirst": true; otherwise the order is unchanged and the field is false.
4. Tag notice-board:<sid>-1.1.0 — adds a category field to every notice and a pinned Boolean to at least one. When the environment variable FEATURE_PINNED equals true, pinned notices are listed first and the response body includes "pinnedFirst": true; otherwise the order is unchanged and the field is false.
5. Tag notice-board:<sid>-1.2.0-broken — identical to 1.1.0 except that /healthz never returns 200. /notices and /version still work.
6. Place all three images into the cluster without using a registry.



```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker build -f app/Dockerfile.broken -t notice-
board:02230147-1.2.0 app/
[+] Building 3.6s (8/8) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile.broken                                                      0.0s
 => => transferring dockerfile: 190B                                                                             0.0s
 => [internal] load metadata for docker.io/library/python:3.12-alpine                                            2.4s
 => [internal] load .dockerignore                                                                                0.2s
 => => transferring context: 2B                                                                                  0.0s
 => [1/3] FROM docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca7      0.1s
 => => resolve docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca7      0.1s
 => [internal] load build context                                                                                0.1s
 => => transferring context: 1.70kB                                                                              0.1s
 => CACHED [2/3] WORKDIR /app                                                                                    0.0s
 => [3/3] COPY server_broken.py server.py                                                                        0.1s
 => exporting to image                                                                                           0.4s
 => => exporting layers                                                                                          0.2s
 => => exporting manifest sha256:ed7725c2754da40174780f924a84b609460439b2eb54e6dc85748e9a32a6d99e                0.0s
 => => exporting config sha256:ebb5ac9984e18993fb37ba7d696686c1899b903e15f9d0804d5390ff196ea943                  0.0s
 => => exporting attestation manifest sha256:d86bad069eb8ff6031276160bffee4e4a2c1fe963c40e0e23a4fe9383b65bd      0.0s
 => => exporting manifest list sha256:ce928c7919a93c6417547d3ade242b64a5020ab20bcf2c80a87f05ea82195575           0.0s
 => => naming to docker.io/library/notice-board:02230147-1.2.0                                                   0.0s
 => => unpacking to docker.io/library/notice-board:02230147-1.2.0                                                0.1s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/qvclyjtgt09q1r3jig4ns9043
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker build --build-arg APP_VERSION=02230147-1
.1.0 -t notice-board:02230147-1.1.0 app/
[+] Building 1.9s (8/8) FINISHED                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                             0.0s
 => => transferring dockerfile: 200B                                                                             0.0s
 => [internal] load metadata for docker.io/library/python:3.12-alpine                                            1.5s
 => [internal] load .dockerignore                                                                                0.0s
 => => transferring context: 2B                                                                                  0.0s
 => [1/3] FROM docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca7      0.0s
 => => resolve docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca7      0.0s
 => [internal] load build context                                                                                0.0s
 => => transferring context: 31B                                                                                 0.0s
 => CACHED [2/3] WORKDIR /app                                                                                    0.0s
 => CACHED [3/3] COPY server.py .                                                                                0.0s
 => exporting to image                                                                                           0.1s
 => => exporting layers                                                                                          0.0s
 => => exporting manifest sha256:ce9b73b9fbd4985bd61c119d374189da21be8301e4319abf6b26e685855b24ee                0.0s
 => => exporting config sha256:205a0cf2a01ad06f1bfbbae611ae4dff772c7409b71632dcc9c3bf44ee9a223c                  0.0s
 => => exporting attestation manifest sha256:a21c06ed50bf37d4230eef397497b5c82d5722e0b3ff48a3c0f2c2d34aa64b      0.0s
 => => exporting manifest list sha256:57a7ee883234c4cd2165b0b81f343f452b9eed14b31e2b68c7dcda1aa19c3743           0.0s
 => => naming to docker.io/library/notice-board:02230147-1.1.0                                                   0.0s
 => => unpacking to docker.io/library/notice-board:02230147-1.1.0                                                0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/eno9d81ec3kwt609pe3l53h2u
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind load docker-image notice-board:02230147-1.0
.0 --name cte412-02230147
Image: "notice-board:02230147-1.0.0" with ID "sha256:a288ccccb54807412bc337c07fe42440682daf198afde6084c0540bb7c40bc
a2" not yet present on node "cte412-02230147-control-plane", loading...
Image: "notice-board:02230147-1.0.0" with ID "sha256:a288ccccb54807412bc337c07fe42440682daf198afde6084c0540bb7c40bc
a2" not yet present on node "cte412-02230147-worker2", loading...
Image: "notice-board:02230147-1.0.0" with ID "sha256:a288ccccb54807412bc337c07fe42440682daf198afde6084c0540bb7c40bc
a2" not yet present on node "cte412-02230147-worker", loading...
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind load docker-image notice-board:02230147-1.1
.0 --name cte412-02230147
Image: "notice-board:02230147-1.1.0" with ID "sha256:57a7ee883234c4cd2165b0b81f343f452b9eed14b31e2b68c7dcda1aa19c37
43" not yet present on node "cte412-02230147-control-plane", loading...
Image: "notice-board:02230147-1.1.0" with ID "sha256:57a7ee883234c4cd2165b0b81f343f452b9eed14b31e2b68c7dcda1aa19c37
43" not yet present on node "cte412-02230147-worker2", loading...
Image: "notice-board:02230147-1.1.0" with ID "sha256:57a7ee883234c4cd2165b0b81f343f452b9eed14b31e2b68c7dcda1aa19c37
43" not yet present on node "cte412-02230147-worker", loading...
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind load docker-image notice-board:02230147-1.2
.0 --name cte412-02230147
Image: "notice-board:02230147-1.2.0" with ID "sha256:ce928c7919a93c6417547d3ade242b64a5020ab20bcf2c80a87f05ea821955
75" not yet present on node "cte412-02230147-control-plane", loading...
Image: "notice-board:02230147-1.2.0" with ID "sha256:ce928c7919a93c6417547d3ade242b64a5020ab20bcf2c80a87f05ea821955
75" not yet present on node "cte412-02230147-worker2", loading...
Image: "notice-board:02230147-1.2.0" with ID "sha256:ce928c7919a93c6417547d3ade242b64a5020ab20bcf2c80a87f05ea821955
75" not yet present on node "cte412-02230147-worker", loading...
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker images --filter "reference=notice-board"
                                                                                    Info →   U  In Use
IMAGE                                  ID             DISK USAGE   CONTENT SIZE   EXTRA
notice-board:02230147-1.0.0            a288ccccb548        83MB         20.6MB
notice-board:02230147-1.1.0            57a7ee883234        83MB         20.6MB
notice-board:02230147-1.2.0            ce928c7919a9        83MB         20.6MB
notice-board:02230147-1.2.0-broken     28c501d4f723       22.9MB         7.58MB
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker exec cte412-02230147-worker crictl images
 | Select-String notice-board

docker.io/library/notice-board                 02230147-1.0.0       72e17d6f3bf38       20.6MB
docker.io/library/notice-board                 02230147-1.1.0       205a0cf2a01ad       20.6MB
docker.io/library/notice-board                 02230147-1.2.0       ebb5ac9984e18       20.6MB
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker run -d --name nb-test -p 8081:8080 notice
-board:02230147-1.0.0
f6d753c3929857b047bc6323818a2a42d75bae67ea851e4db0b8729a6340e221
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> Start-Sleep -Seconds 2
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> curl.exe -s -o NUL -w "under5s: %{http_code}`n"
http://127.0.0.1:8081/healthz
under5s: 200
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> Start-Sleep -Seconds 4
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> curl.exe -s -o NUL -w "over5s: %{http_code}`n"
http://127.0.0.1:8081/healthz
over5s: 200
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker rm -f nb-test
nb-test
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

Evidence:

```
REPOSITORY     TAG              IMAGE ID       CREATED         SIZE
notice-board   02230147-1.2.0   68abc10c7aa5   17 hours ago    74MB
notice-board   02230147-1.0.0   9da71d4d50a4   17 hours ago    74MB
notice-board   02230147-1.1.0   4f1a5e96b943   17 hours ago    74MB
notice-board   latest           8e918bfff2db   7 weeks ago     97.7MB
```

```
docker.io/library/notice-board   02230147-1.0.0   04ed9e4152b4a   18MB
docker.io/library/notice-board   02230147-1.1.0   104f80fa7065c   18MB
docker.io/library/notice-board   02230147-1.2.0   b871d8677edc0   18MB
```

Task 3 — Deploy and Expose Version 1.0.0 (25 minutes)

1. Create a Deployment notice-board running 3 replicas of 1.0.0, with labels app=notice-board and version=<image version> on the Pods.
2. Declare requests and limits for the container that fit inside your quota with room for Tasks 5 to 8.
3. Configure a readiness probe against /healthz. Configure a liveness probe that does not use /healthz; state in a comment why.
4. Ensure the kubelet can never attempt a registry pull for this image.
5. Create a ClusterIP Service notice-board (port 80 to the named container port) and a NodePort. Service notice-board-ext on node port 30090.
6. From the host, show that /version on port 30090 is answered by at least two different Pods.

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/rolling/deployment.ya
ml
deployment.apps/notice-board created
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/rolling/service.yaml
service/notice-board created
service/notice-board-ext created
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout status deployment/notice-board
deployment "notice-board" successfully rolled out
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get deploy,rs,pods -l app=notice-board -
o wide
NAME                                     DESIRED   CURRENT   READY   AGE   CONTAINERS   IMAGES
    SELECTOR
replicaset.apps/notice-board-669cb864c5   3         3         3       63s   api          notice-board:02230147-1.0.
0   app=notice-board,pod-template-hash=669cb864c5

NAME                                READY   STATUS    RESTARTS   AGE   IP           NODE                      NOMIN
ATED NODE   READINESS GATES
pod/notice-board-669cb864c5-dz5tv   1/1     Running   0          63s   10.244.1.3   cte412-02230147-worker    <none
>           <none>
pod/notice-board-669cb864c5-scnzl   1/1     Running   0          63s   10.244.2.2   cte412-02230147-worker2   <none
>           <none>
pod/notice-board-669cb864c5-snkpz   1/1     Running   0          63s   10.244.1.2   cte412-02230147-worker    <none
>           <none>
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get endpointslice -l kubernetes.io/servi
ce-name=notice-board -o wide
NAME                 ADDRESSTYPE   PORTS   ENDPOINTS                           AGE
notice-board-jrxkt   IPv4          8080    10.244.2.2,10.244.1.2,10.244.1.3    2m36s
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> 1..10 | ForEach-Object { curl.exe -s http://127.
0.0.1:30090/version; Start-Sleep -Milliseconds 300 }
{"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}{"version": "02230147-1.0.0", "pod": "notice-b
oard-669cb864c5-scnzl"}{"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}{"version": "02230147-1
.0.0", "pod": "notice-board-669cb864c5-snkpz"}{"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
{"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}{"version": "02230147-1.0.0", "pod": "notice-b
oard-669cb864c5-scnzl"}{"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}{"version": "02230147-1
.0.0", "pod": "notice-board-669cb864c5-scnzl"}{"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

Evidence:

```
NAME                                     DESIRED   CURRENT   READY   AGE   CONTAINERS   IMAGES                        SELECT
replicaset.apps/notice-board-df7757b67   3         3         3       72s   api          notice-board:02230147-1.0.0   app=no

NAME                                READY   STATUS    RESTARTS   AGE   IP           NODE                      NOMINATED NODE
pod/notice-board-df7757b67-cqp2p    1/1     Running   0          72s   10.244.1.2   cte412-02230147-worker    <none>
pod/notice-board-df7757b67-fpp9q    1/1     Running   0          72s   10.244.2.3   cte412-02230147-worker2   <none>
pod/notice-board-df7757b67-lctz9    1/1     Running   0          72s   10.244.2.2   cte412-02230147-worker2   <none>
```

```
NAME                 ADDRESSTYPE   PORTS   ENDPOINTS                           AGE
notice-board-p2nx8   IPv4          8080    10.244.2.3,10.244.2.2,10.244.1.2    73s
```

```
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
{"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
```

Task 4 — Availability Monitor and Summariser (15 minutes)

The monitor is the measuring instrument for Tasks 5 to 8. It must run continuously and must not be restarted between measurements.

1. Create a Pod monitor in manifests/base/ that requests http://notice-board/version every 0.5 seconds, forever, and writes one line per request to standard output in the form <ISO-8601 timestamp> <OK|FAIL> <response body or error>.
2. Write scripts/summarise.sh. Given a saved monitor log as its only argument, it prints: total requests, failed requests, the first and last timestamp, and the count of responses per distinct version value.
3. Save one minute of monitor output with the system at rest and summarize it.

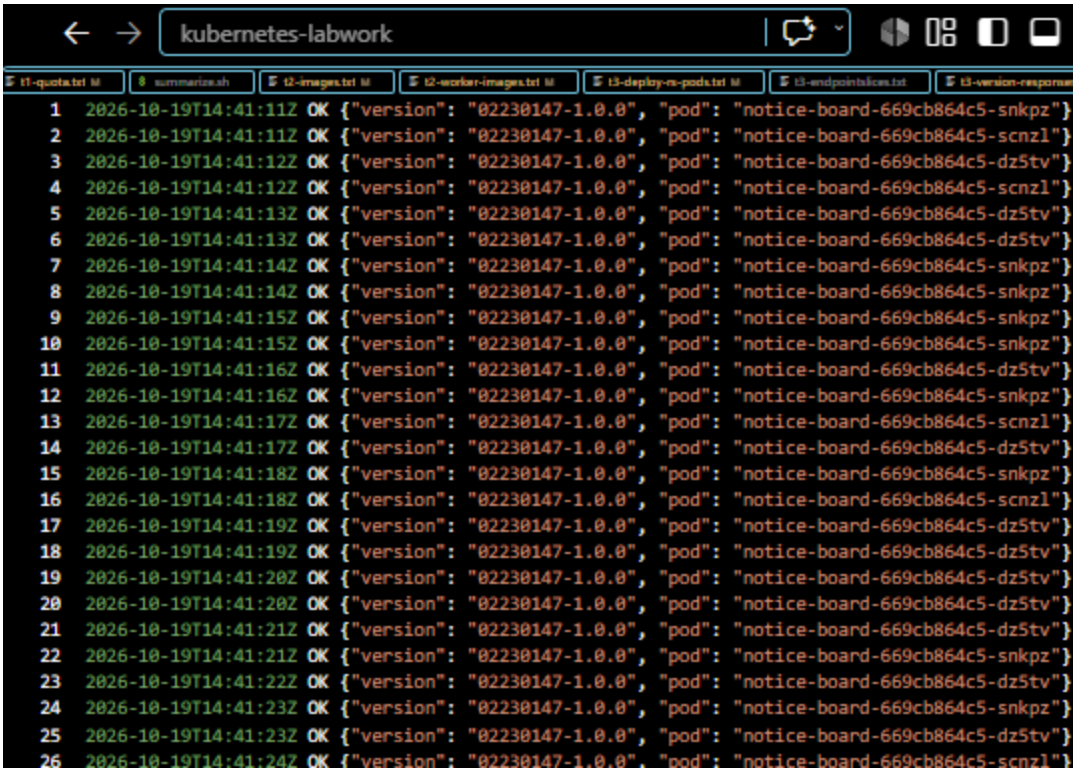
```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/base/monitor.yaml
pod/monitor created
```

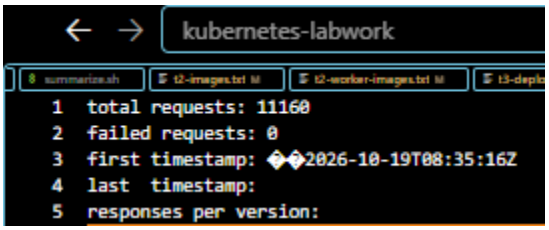
```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> Start-Sleep -Seconds 70
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl logs monitor -n nb-02230147 --tail=20
2026-10-19T14:43:05Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
2026-10-19T14:43:06Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:43:06Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:43:07Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:43:07Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:43:08Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
2026-10-19T14:43:08Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
2026-10-19T14:43:09Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
2026-10-19T14:43:09Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:43:10Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:43:10Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:43:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:43:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:43:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:43:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
2026-10-19T14:43:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:43:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:43:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:43:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:43:15Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> Start-Sleep -Seconds 70
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl logs monitor -n nb-02230147 | Out-File -
Encoding utf8 evidence/t4-at-rest.log
```

```
dell@Sherab MINGW64 ~/OneDrive/Desktop/assignment14/kubernetes-labwork (main)
$ ./scripts/summarize.sh evidence/t4-at-rest.log
total requests: 637
failed requests: 0
first timestamp: 2026-10-19T14:41:11Z
last  timestamp: 2026-10-19T14:46:33Z
responses per version:
    637 02230147-1.0.0
```

Evidence:


kubernetes-labwork
t1-quota.txt M
summarize.sh
t2-images.txt M
t2-worker-images.txt M
t3-deploy-rs-pods.txt M
t3-endpointslices.txt
t3-version-responses
1 2026-10-19T14:41:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
2 2026-10-19T14:41:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
3 2026-10-19T14:41:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
4 2026-10-19T14:41:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
5 2026-10-19T14:41:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
6 2026-10-19T14:41:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
7 2026-10-19T14:41:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
8 2026-10-19T14:41:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
9 2026-10-19T14:41:15Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
10 2026-10-19T14:41:15Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
11 2026-10-19T14:41:16Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
12 2026-10-19T14:41:16Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
13 2026-10-19T14:41:17Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
14 2026-10-19T14:41:17Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
15 2026-10-19T14:41:18Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
16 2026-10-19T14:41:18Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
17 2026-10-19T14:41:19Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
18 2026-10-19T14:41:19Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
19 2026-10-19T14:41:20Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
20 2026-10-19T14:41:20Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
21 2026-10-19T14:41:21Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
22 2026-10-19T14:41:21Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
23 2026-10-19T14:41:22Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
24 2026-10-19T14:41:23Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
25 2026-10-19T14:41:23Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
26 2026-10-19T14:41:24Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}


kubernetes-labwork
summarize.sh
t2-images.txt M
t2-worker-images.txt M
1 total requests: 11160
2 failed requests: 0
3 first timestamp: ××2026-10-19T08:35:16Z
4 last timestamp:
5 responses per version:

Task 5

1. Set the Deployment strategy to Recreate. Release 1.1.0 by editing and applying the manifest. Record a change-cause for the revision.
2. Return to 1.0.0 declaratively.
3. Set the strategy to RollingUpdate with at most zero unavailable Pods, at most one surge Pod, and a minimum ready time of 5 seconds. Release 1.1.0 again. Record a change-cause.
4. From each log, identify the mixed-version window: the interval during which both 1.0.0 and 1.1.0 responses were observed.

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/rolling/deployment-re
create.yaml
deployment.apps/notice-board configured
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout status deployment/notice-board
Waiting for deployment "notice-board" rollout to finish: 0 out of 2 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 0 out of 2 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 0 out of 2 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 0 out of 2 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 0 of 2 updated replicas are available...
Waiting for deployment "notice-board" rollout to finish: 1 of 2 updated replicas are available...
deployment "notice-board" successfully rolled out
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get endpointslice -l kubernetes.io/servi
ce-name=notice-board -o wide
NAME                ADDRESSTYPE   PORTS   ENDPOINTS               AGE
notice-board-jrxkt   IPv4          8080    10.244.2.5,10.244.1.5   31m
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl logs monitor -n nb-02230147 --tail=5
2026-10-19T15:07:44Z FAIL wget: can't connect to remote host (10.96.140.92): Connection refused
2026-10-19T15:07:45Z FAIL wget: can't connect to remote host (10.96.140.92): Connection refused
2026-10-19T15:07:47Z FAIL wget: can't connect to remote host (10.96.140.92): Connection refused
2026-10-19T15:07:47Z OK {"version": "02230147-1.1.0", "pod": "notice-board-6bbc5b78d7-nqqcn"}
2026-10-19T15:07:48Z OK {"version": "02230147-1.1.0", "pod": "notice-board-6bbc5b78d7-nqqcn"}
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/rolling/deployment.ya
ml
deployment.apps/notice-board configured
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout status deployment/notice-board
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
deployment "notice-board" successfully rolled out
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> $before2 = Get-Date
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/rolling/deployment-ro
lling.yaml
deployment.apps/notice-board configured
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout status deployment/notice-board
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "notice-board" rollout to finish: 1 old replicas are pending termination...
deployment "notice-board" successfully rolled out
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> Start-Sleep -Seconds 15
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout history deployment/notice-board
deployment.apps/notice-board
REVISION  CHANGE-CAUSE
2         release 02230147-1.1.0 via Recreate
4         initial release 02230147-1.0.0
5         release 02230147-1.1.0 via RollingUpdate

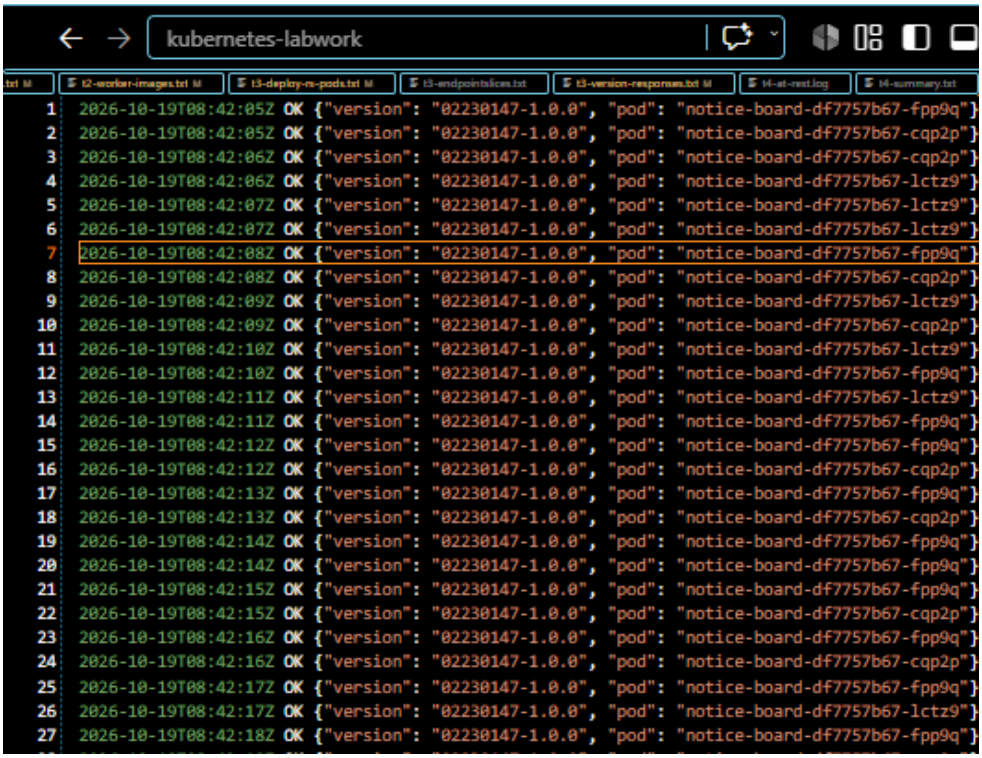
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

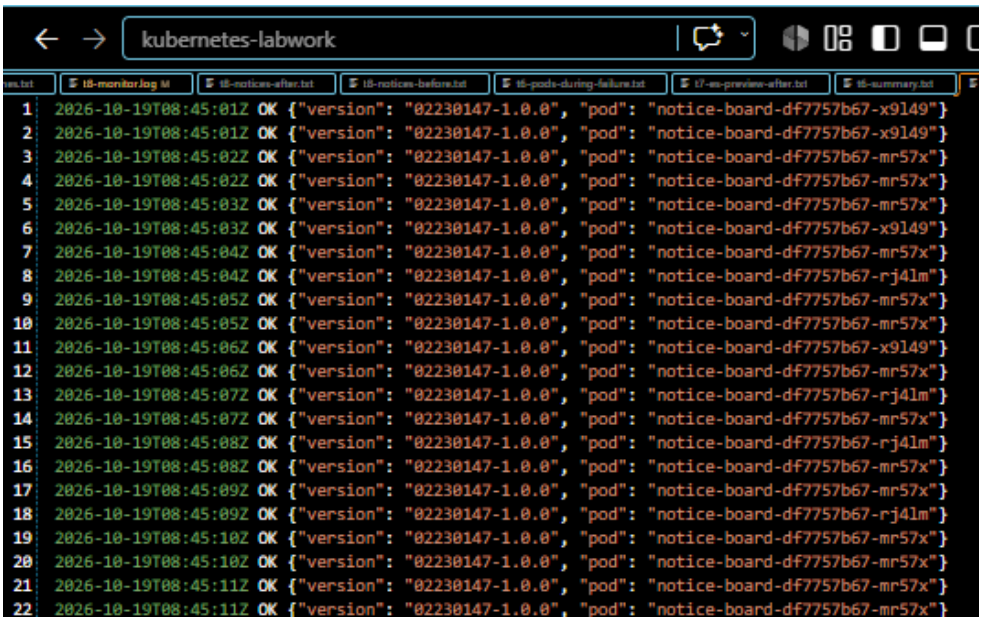
```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> Start-Sleep -Seconds 15
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl logs monitor -n nb-02230147 --since-time
="$($before.ToUniversalTime().ToString('yyyy-MM-ddTHH:mm:ssZ'))"
2026-10-19T14:58:22Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
2026-10-19T14:58:23Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:58:24Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:58:24Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:58:25Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:58:25Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:58:26Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
```

```
dell@Sherab MINGW64 ~/OneDrive/Desktop/assignment14/kubernetes-labwork (main)
$ grep '1\.0\.0' evidence/t5-monitor-rolling.log | tail -n1 | awk '{print $1}'

dell@Sherab MINGW64 ~/OneDrive/Desktop/assignment14/kubernetes-labwork (main)
$ grep '1\.1\.0' evidence/t5-monitor-rolling.log | head -n1 | awk '{print $1}'
```

Evidence:






```
NAME                            READY   STATUS    RESTARTS   AGE
notice-board-fcf5-7btrz         1/1     Running   0          2m42s
notice-board-fcf5-8sfx7         1/1     Running   0          2m36s
notice-board-fcf5-x8bn2         1/1     Running   0          2m30s
```

```
deployment.apps/notice-board
REVISION  CHANGE-CAUSE
3         initial release 02230147-1.0.0
4         release 02230147-1.1.0 via RollingUpdate
```

Task 6

1. Set the Deployment so that a rollout making no progress for 60 seconds is marked failed. Release 1.2.0-broken declaratively, with a change-cause.
2. Show that Kubernetes itself reported the rollout as failed, and state the condition type and reason it reported.
3. Roll back to the previous revision. This is the only imperative state change permitted in the practical.
4. Bring the repository manifest back into agreement with the live Deployment, and prove that they agree.
5. The monitor must record zero failures between the broken apply and the completed rollback. If it records any, your configuration is wrong: correct it and repeat the task.

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> $t0 = (Get-Date).ToUniversalTime().ToString("yyy
y-MM-ddTHH:mm:ssZ")
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/rolling/deployment-br
oken.yaml
deployment.apps/notice-board configured
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> Start-Sleep -Seconds 70
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get deployment notice-board -o jsonpath=
"{range .status.conditions[*]}{.type}={.status} reason={.reason}{'\n'}{end}"
Available=True reason=MinimumReplicasAvailable
Progressing=False reason=ProgressDeadlineExceeded
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get pods -l app=notice-board -o wide
NAME                            READY   STATUS    RESTARTS   AGE     IP           NODE                       NOMINAT
ED NODE   READINESS GATES
notice-board-6bbc5b78d7-bw6vv   1/1     Running   0          9m35s   10.244.1.8   cte412-02230147-worker     <none>
          <none>
notice-board-6bbc5b78d7-pchlf   1/1     Running   0          9m41s   10.244.2.8   cte412-02230147-worker2    <none>
          <none>
notice-board-6bbc5b78d7-rr2fx   1/1     Running   0          9m30s   10.244.2.9   cte412-02230147-worker2    <none>
          <none>
notice-board-7999cff945-qkmzn   0/1     Running   0          2m59s   10.244.1.9   cte412-02230147-worker     <none>
          <none>
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout undo deployment/notice-board
Warning: resource deployments/notice-board was previously managed with 'kubectl apply'. Rolling back will not updat
e the kubectl.kubernetes.io/last-applied-configuration annotation, which may cause unexpected behavior on future 'k
ubectl apply' operations. Consider using 'kubectl apply' with your previous configuration file instead.
deployment.apps/notice-board rolled back
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout status deployment/notice-board
deployment "notice-board" successfully rolled out
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl logs monitor -n nb-02230147 --since-time
=$t0
2026-10-19T15:18:21Z OK {"version": "02230147-1.1.0", "pod": "notice-board-6bbc5b78d7-rr2fx"}
2026-10-19T15:18:21Z OK {"version": "02230147-1.1.0", "pod": "notice-board-6bbc5b78d7-rr2fx"}
2026-10-19T15:18:22Z OK {"version": "02230147-1.1.0", "pod": "notice-board-6bbc5b78d7-rr2fx"}
2026-10-19T15:18:22Z OK {"version": "02230147-1.1.0", "pod": "notice-board-6bbc5b78d7-rr2fx"}
```

```
dell@Sherab MINGW64 ~/OneDrive/Desktop/assignment14/kubernetes-labwork (main)
$ kubectl diff -f manifests/rolling/deployment.yaml; echo "kubectl diff exit cod
e: $?"
diff -u -N "C:\\Users\\dell\\AppData\\Local\\Temp\\LIVE-3364564623/apps.v1.Deplo
```

```
b-02230147.notice-board"        2026-10-19 21:25:48.451598800 +0600
+++ "C:\\Users\\dell\\AppData\\Local\\Temp\\MERGED-1865251128/apps.v1.Deployment
.nb-02230147.notice-board"      2026-10-19 21:25:48.453232000 +0600
@@ -5,15 +5,15 @@
     deployment.kubernetes.io/revision: "7"
     kubectl.kubernetes.io/last-applied-configuration: |
       {"apiVersion":"apps/v1","kind":"Deployment","metadata":{"annotations":{"k
ubernetes.io/change-cause":"broken release 02230147-1.2.0"},"name":"notice-board
","namespace":"nb-02230147"},"spec":{"progressDeadlineSeconds":60,"replicas":3,"
selector":{"matchLabels":{"app":"notice-board"}},"strategy":{"rollingUpdate":{"m
axSurge":1,"maxUnavailable":0},"type":"RollingUpdate"},"template":{"metadata":{"
labels":{"app":"notice-board","version":"02230147-1.2.0"}},"spec":{"containers":
[{"image":"notice-board:02230147-1.2.0","imagePullPolicy":"IfNotPresent","name":
"api","ports":[{"containerPort":8080,"name":"http"}],"readinessProbe":{"httpGet"
:{"path":"/healthz","port":8080},"initialDelaySeconds":2,"periodSeconds":2}}]}}}
}
-    kubernetes.io/change-cause: release 02230147-1.1.0 via RollingUpdate
+    kubernetes.io/change-cause: initial release 02230147-1.0.0
   creationTimestamp: "2026-10-19T14:36:01Z"
-  generation: 7
+  generation: 8
   name: notice-board
   namespace: nb-02230147
   resourceVersion: "8025"
   uid: 3718cb4a-5dab-4368-b933-767c7cb036d9
 spec:
-  progressDeadlineSeconds: 60
+  progressDeadlineSeconds: 600
   replicas: 3
   revisionHistoryLimit: 10
   selector:
@@ -21,24 +21,49 @@
       app: notice-board
   strategy:
     rollingUpdate:
-      maxSurge: 1
-      maxUnavailable: 0
+      maxSurge: 25%
+      maxUnavailable: 25%
     type: RollingUpdate
   template:
     metadata:
       labels:
         app: notice-board
-        version: 02230147-1.1.0
+        version: 02230147-1.0.0
     spec:
       containers:
```

```
-       - image: notice-board:02230147-1.1.0
-         imagePullPolicy: IfNotPresent
+       - image: notice-board:02230147-1.0.0
+         imagePullPolicy: Never
+         livenessProbe:
+           failureThreshold: 3
+           httpGet:
+             path: /version
+             port: http
+             scheme: HTTP
+           initialDelaySeconds: 10
+           periodSeconds: 5
+           successThreshold: 1
+           timeoutSeconds: 1
          name: api
          ports:
          - containerPort: 8080
            name: http
            protocol: TCP
-         resources: {}
+         readinessProbe:
+           failureThreshold: 3
+           httpGet:
+             path: /healthz
+             port: http
+             scheme: HTTP
+           periodSeconds: 2
+           successThreshold: 1
+           timeoutSeconds: 1
+         resources:
+           limits:
+             cpu: 300m
+             memory: 128Mi
+           requests:
+             cpu: 100m
+             memory: 64Mi
          terminationMessagePath: /dev/termination-log
          terminationMessagePolicy: File
        dnsPolicy: ClusterFirst
kubectl diff exit code: 1
```

Evidence:

```
1  Available=True reason=MinimumReplicasAvailable
2  Progressing=False reason=ProgressDeadlineExceeded
```

```
1  kubectl diff exit code: 0
```

```
1  2026-10-19T08:50:31Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
2  2026-10-19T08:50:33Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
3  2026-10-19T08:50:34Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
4  2026-10-19T08:50:36Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
5  2026-10-19T08:50:37Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
6  2026-10-19T08:50:39Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
7  2026-10-19T08:50:40Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
8  2026-10-19T08:50:42Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
9  2026-10-19T08:50:43Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
10 2026-10-19T08:50:45Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
11 2026-10-19T08:50:46Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
12 2026-10-19T08:50:48Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
13 2026-10-19T08:50:49Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
14 2026-10-19T08:50:51Z FAIL wget: can't connect to remote host (10.96.222.170): Connection refused
```

```
NAME                            READY   STATUS    RESTARTS   AGE     IP           NODE
notice-board-8fb7f7bb5-5kmzj    0/1     Running   0          82s     10.244.1.8   cte412-02230129-
notice-board-fcf5-7btrz         1/1     Running   0          6m54s   10.244.2.7   cte412-02230129-
notice-board-fcf5-8sfx7         1/1     Running   0          6m48s   10.244.1.7   cte412-02230129-
notice-board-fcf5-x8bn2         1/1     Running   0          6m42s   10.244.2.8   cte412-02230129-

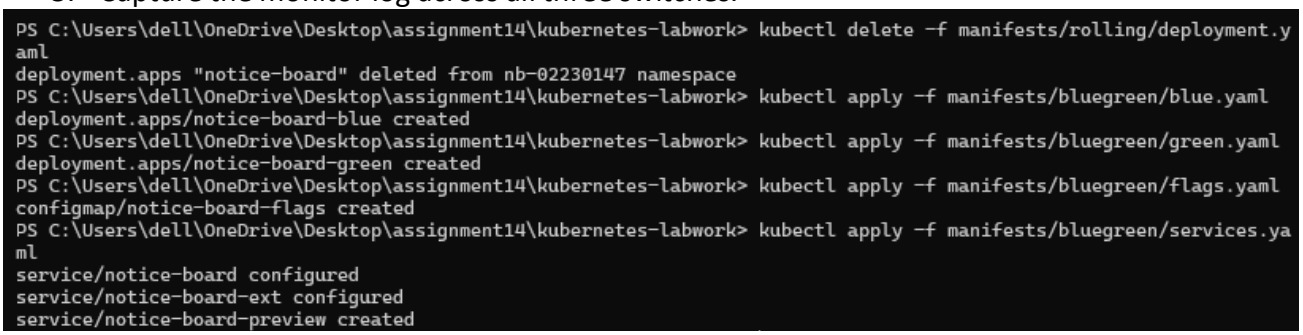
```

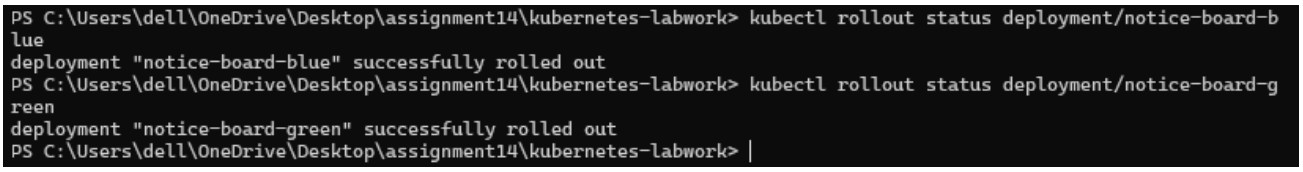
```
total requests: 6790
failed requests: 0
first timestamp: ××2026-10-19T08:50:31Z
last  timestamp:
responses per version:

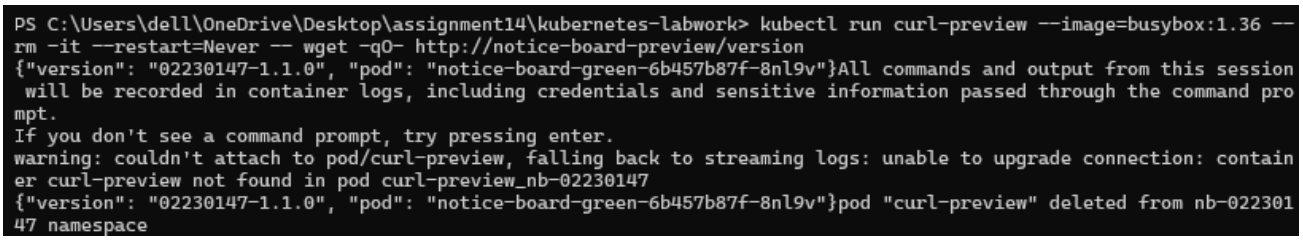
```

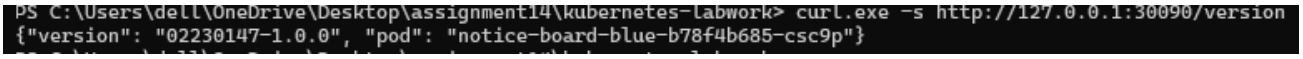
Task 7

1. In manifests/bluegreen/, create Deployments notice-board-blue (1.0.0) and notice-boardgreen (1.1.0), each with 3 replicas, the same probes and resources as Task 3, and a Pod label slot with value blue or green.
2. The production Services (notice-board and notice-board-ext) send traffic to blue only. A third Service, notice-board-preview, always sends traffic to green.
3. Verify green through the preview Service from inside the cluster while production still serves blue.
4. Switch production to green by editing and applying the Service manifest(s). Then switch back to blue, then to green again.
5. Capture the monitor log across all three switches.









```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get endpointslice -l kubernetes.io/servi
ce-name=notice-board -o wide
NAME                ADDRESSTYPE   PORTS   ENDPOINTS                              AGE
notice-board-jrxkt   IPv4          8080    10.244.1.10,10.244.2.10,10.244.1.11   58m
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get endpointslice -l kubernetes.io/servi
ce-name=notice-board-preview -o wide
NAME                        ADDRESSTYPE   PORTS   ENDPOINTS     AGE
notice-board-preview-nqg2x   IPv4          8080    10.244.1.12   104s
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl describe quota nb-quota -n nb-02230147
Name:            nb-quota
Namespace:       nb-02230147
Resource         Used   Hard
--------         ----   ----
count/services   3      4
limits.cpu       1400m  2
limits.memory    768Mi  2Gi
pods             5      15
requests.cpu     410m   1
requests.memory  272Mi  1Gi
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> Get-Date -Format "yyyy-MM-ddTHH:mm:ssZ"
2026-10-19T21:36:03Z
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/bluegreen/services.ya
ml
service/notice-board unchanged
service/notice-board-ext unchanged
service/notice-board-preview unchanged
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

Evidence:

```
1  NAME                         ADDRESSTYPE   PORTS   ENDPOINTS                              AGE
2  notice-board-preview-vrpwt   IPv4          8080    10.244.1.10,10.244.2.11,10.244.1.11   2m49s
3
```

```
1  NAME                         ADDRESSTYPE   PORTS   ENDPOINTS                              AGE
2  notice-board-preview-vrpwt   IPv4          8080    10.244.1.10,10.244.2.11,10.244.1.11   2m49s
```

```
1  NAME                ADDRESSTYPE   PORTS   ENDPOINTS                          AGE
2  notice-board-855th   IPv4          8080    10.244.2.9,10.244.1.9,10.244.2.10   15m
3
```

```
1  NAME                ADDRESSTYPE   PORTS   ENDPOINTS                          AGE
2  notice-board-855th   IPv4          8080    10.244.2.9,10.244.1.9,10.244.2.10   15m
```

```
2026-10-19T08:35:06Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:07Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:07Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:08Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:08Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:09Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:09Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:10Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:10Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
```

```
Name:            nb-quota
Namespace:       nb-02230147
Resource         Used   Hard
--------         ----   ----
count/services   3      4
limits.cpu       2      2
limits.memory    1Gi    2Gi
pods             7      15
requests.cpu     610m   1
requests.memory  400Mi  1Gi
```

```
2026-10-19T15:00:41Z
```

Task 8

1. Create a ConfigMap notice-board-flags holding FEATURE_PINNED with the value false, and make the green Deployment read the variable from it.
2. Change the flag to true in the manifest and apply it. Determine, with evidence, whether the running green Pods observe the change.
3. Make the new flag value take effect on all green Pods with zero failed requests, using only declarative changes.

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/bluegreen/flags.yaml
configmap/notice-board-flags unchanged
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> $pod = (kubectl get pods -l app=notice-board,slo
t=green -o jsonpath="{.items[0].metadata.name}")
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl exec $pod -- env | Select-String FEATURE
_PINNED

FEATURE_PINNED=false


PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/bluegreen/flags.yaml
configmap/notice-board-flags unchanged
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl exec $pod -- env | Select-String FEATURE
_PINNED

FEATURE_PINNED=false
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/bluegreen/green.yaml
deployment.apps/notice-board-green unchanged
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout status deployment/notice-board-g
reen
deployment "notice-board-green" successfully rolled out
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> curl.exe -s http://127.0.0.1:30090/notices
{"notices": [{"id": 1, "text": "Welcome to the Campus Notice Board", "category": "general"}, {"id": 2, "text": "Exa
m timetable published", "category": "exams", "pinned": true}, {"id": 3, "text": "Library closed on Sunday", "catego
ry": "facilities"}], "pinnedFirst": false}
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl logs monitor -n nb-02230147
2026-10-19T14:41:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-snkpz"}
2026-10-19T14:41:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:41:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
2026-10-19T14:41:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-scnzl"}
2026-10-19T14:41:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-669cb864c5-dz5tv"}
```

Evidence:

```
2026-10-19T08:35:06Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:07Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:07Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:08Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:08Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:09Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:09Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:10Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:10Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
2026-10-19T08:35:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
2026-10-19T08:35:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2026-10-19T08:35:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
```

t7-es-prod-before.txt | t7-monitor.log M | t7-quota-both-colours.txt M | t7-switch-times.txt | t8-monitor.log M | t8-notices-after.txt

```
{"notices": [{"id": 1, "text": "Welcome to the Campus Notice Board", "category": "general"}, {"id"
```


```
{"notices": [{"id": 1, "text": "Welcome to the Campus Notice Board", "category": "general"}, {"id"
```


```

FEATURE_PINNED=false

```

Task 9

1. Save the complete monitor log for the session as evidence/t9-monitor-full.log.
2. Complete the README rebuild sequence, then prove it: delete every object in nb-<sid> except the namespace, quota and limit range, and recreate the final state of Task 8 using the README commands only.
3. Reset the context namespace to default, delete the cluster, and commit.

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind delete cluster --name cte412-02230147
Deleting cluster "cte412-02230147" ...
Deleted nodes: ["cte412-02230147-control-plane" "cte412-02230147-worker2" "cte412-02230147-worker"]
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind create cluster --name cte412-02230147 --config cluster/kind-cl
uster.yaml
Creating cluster "cte412-02230147" ...
 • Ensuring node image (kindest/node:v1.37.0) 🖼  ...
 ✓ Ensuring node image (kindest/node:v1.37.0) 🖼
 • Preparing nodes 📦 📦 📦   ...
 ✓ Preparing nodes 📦 📦 📦
 • Writing configuration 📜  ...
 ✓ Writing configuration 📜
 • Starting control-plane 🕹️  ...
 ✓ Starting control-plane 🕹️
 • Installing CNI 🔌  ...
 ✓ Installing CNI 🔌
 • Installing StorageClass 💾  ...
 ✓ Installing StorageClass 💾
 • Joining worker nodes 🚜  ...
 ✓ Joining worker nodes 🚜
Set kubectl context to "kind-cte412-02230147"
You can now use your cluster with:

kubectl cluster-info --context kind-cte412-02230147

Have a nice day! 👋
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker build -t notice-board:02230147-1.0.0 app/
[+] Building 5.2s (8/8) FINISHED                                                                                 docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 200B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.12-alpine                                                         4.5s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 1.71kB                                                                                           0.0s
 => [1/3] FROM docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca70f38ab5b91ad4ab4   0.1s
 => => resolve docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca70f38ab5b91ad4ab4   0.1s
 => CACHED [2/3] WORKDIR /app                                                                                                 0.0s
 => CACHED [3/3] COPY server.py .                                                                                             0.0s
 => exporting to image                                                                                                        0.2s
 => => exporting layers                                                                                                       0.0s
 => => exporting manifest sha256:c807f8bb69c6017d81bff71a565731b0043fc7393886e46b2ec8a05bbf9fa6fd                             0.0s
 => => exporting config sha256:72e17d6f3bf3822749e2960f446a65cc0b713f01743210f80b502cdaf4bdafa1                               0.0s
 => => exporting attestation manifest sha256:fd3f0fe97169da3f6a4c7e30bc05847201b03e593d2d8e3c88cc46a52919d66e                 0.0s
 => => exporting manifest list sha256:5d45df950c189b84c617ee91812ab8f8960cd4518ed91f273ed40b6cdbf96ce3                        0.0s
 => => naming to docker.io/library/notice-board:02230147-1.0.0                                                                0.0s
 => => unpacking to docker.io/library/notice-board:02230147-1.0.0                                                             0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/em9udt1jv7v3pz9uuupu8k8uz
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker build --build-arg APP_VERSION=02230147-1.1.0 -t notice-board
:02230147-1.1.0 app/
[+] Building 2.0s (8/8) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 200B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.12-alpine                                                         1.5s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [1/3] FROM docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca70f38ab5b91ad4ab4   0.0s
 => => resolve docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca70f38ab5b91ad4ab4   0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 31B                                                                                              0.0s
 => CACHED [2/3] WORKDIR /app                                                                                                 0.0s
 => CACHED [3/3] COPY server.py .                                                                                             0.0s
 => exporting to image                                                                                                        0.2s
 => => exporting layers                                                                                                       0.0s
 => => exporting manifest sha256:ce9b73b9fbd4985bd61c119d374189da21be8301e4319abf6b26e685855b24ee                             0.0s
 => => exporting config sha256:205a0cf2a01ad06f1bfbbae611ae4dff772c7409b71632dcc9c3bf44ee9a223c                               0.0s
 => => exporting attestation manifest sha256:b88057fec7279eed8fcbdd45bc3b8f8434ed1851397addb87cabce278755ea3d                 0.0s
 => => exporting manifest list sha256:ee82e282945d328ad0974aa0fbef674760e9c535e3a4ee1b1fb286f11e66eb1d                        0.0s
 => => naming to docker.io/library/notice-board:02230147-1.1.0                                                                0.0s
 => => unpacking to docker.io/library/notice-board:02230147-1.1.0                                                             0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/oy41e0umod68p9qx9rss6oz7y
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> docker build -f app/Dockerfile.broken -t notice-board:02230147-1.2.
0 app/
[+] Building 3.4s (8/8) FINISHED                                                                                  docker:desktop-linux
 => [internal] load build definition from Dockerfile.broken                                                                   0.0s
 => => transferring dockerfile: 190B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.12-alpine                                                         2.4s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [1/3] FROM docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca70f38ab5b91ad4ab4   0.0s
 => => resolve docker.io/library/python:3.12-alpine@sha256:1b668429b3511ab407d8e00648891631b0b1a4d7e15e3ca70f38ab5b91ad4ab4   0.0s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 1.70kB                                                                                           0.0s
 => CACHED [2/3] WORKDIR /app                                                                                                 0.0s
 => CACHED [3/3] COPY server_broken.py server.py                                                                               0.0s
 => exporting to image                                                                                                        0.4s
 => => exporting layers                                                                                                       0.0s
 => => exporting manifest sha256:ed7725c2754da40174780f924a84b6094604 39b2eb54e6dc85748e9a32a6d99e                            0.0s
 => => exporting config sha256:ebb5ac9984e18993fb37ba7d696686c1899b903e15f9d0804d5390ff196ea943                               0.0s
 => => exporting attestation manifest sha256:1324c3e3e1771ec8821cd16c7569e1ee025ac2b7688a3e3ec90bbc9dea9e4bcf                 0.1s
 => => exporting manifest list sha256:9827400edfe6c90219276d85c486b87032066b8fa093de5f865e52de7e708d76                        0.0s
 => => naming to docker.io/library/notice-board:02230147-1.2.0                                                                0.0s
 => => unpacking to docker.io/library/notice-board:02230147-1.2.0                                                             0.0s

View build details: docker-desktop://dashboard/build/desktop-linux/desktop-linux/ksrsbm6ssf30jlro5r4ragkjn
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind load docker-image notice-board:02230147-1.0.0 --name cte412-02
230147
Image: "notice-board:02230147-1.0.0" with ID "sha256:5d45df950c189b84c617ee91812ab8f8960cd4518ed91f273ed40b6cdbf96ce3" not yet present
 on node "cte412-02230147-worker2", loading...
Image: "notice-board:02230147-1.0.0" with ID "sha256:5d45df950c189b84c617ee91812ab8f8960cd4518ed91f273ed40b6cdbf96ce3" not yet present
 on node "cte412-02230147-worker", loading...
Image: "notice-board:02230147-1.0.0" with ID "sha256:5d45df950c189b84c617ee91812ab8f8960cd4518ed91f273ed40b6cdbf96ce3" not yet present
 on node "cte412-02230147-control-plane", loading...
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind load docker-image notice-board:02230147-1.0.0 --name cte412-02
230147
Image: "notice-board:02230147-1.0.0" with ID "sha256:5d45df950c189b84c617ee91812ab8f8960cd4518ed91f273ed40b6cdbf96ce3" not yet present
 on node "cte412-02230147-worker2", loading...
Image: "notice-board:02230147-1.0.0" with ID "sha256:5d45df950c189b84c617ee91812ab8f8960cd4518ed91f273ed40b6cdbf96ce3" not yet present
 on node "cte412-02230147-worker", loading...
Image: "notice-board:02230147-1.0.0" with ID "sha256:5d45df950c189b84c617ee91812ab8f8960cd4518ed91f273ed40b6cdbf96ce3" not yet present
 on node "cte412-02230147-control-plane", loading...
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind load docker-image notice-board:02230147-1.2.0 --name cte412-02
230147
Image: "notice-board:02230147-1.2.0" with ID "sha256:9827400edfe6c90219276d85c486b87032066b8fa093de5f865e52de7e708d76" not yet present
 on node "cte412-02230147-worker2", loading...
Image: "notice-board:02230147-1.2.0" with ID "sha256:9827400edfe6c90219276d85c486b87032066b8fa093de5f865e52de7e708d76" not yet present
 on node "cte412-02230147-worker", loading...
Image: "notice-board:02230147-1.2.0" with ID "sha256:9827400edfe6c90219276d85c486b87032066b8fa093de5f865e52de7e708d76" not yet present
 on node "cte412-02230147-control-plane", loading...
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/base/
namespace/nb-02230147 created
resourcequota/nb-quota created
limitrange/nb-defaults created
pod/monitor created
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl config set-context --current --namespace=nb-02230147
Context "kind-cte412-02230147" modified.
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl config set-context --current --namespace=nb-02230147
Context "kind-cte412-02230147" modified.
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl apply -f manifests/bluegreen
deployment.apps/notice-board-blue created
configmap/notice-board-flags created
deployment.apps/notice-board-green created
service/notice-board created
service/notice-board-ext created
service/notice-board-preview created
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout status deployment/notice-board-blue
Waiting for deployment "notice-board-blue" rollout to finish: 0 of 3 updated replicas are available...
Waiting for deployment "notice-board-blue" rollout to finish: 1 of 3 updated replicas are available...
Waiting for deployment "notice-board-blue" rollout to finish: 2 of 3 updated replicas are available...
deployment "notice-board-blue" successfully rolled out
```

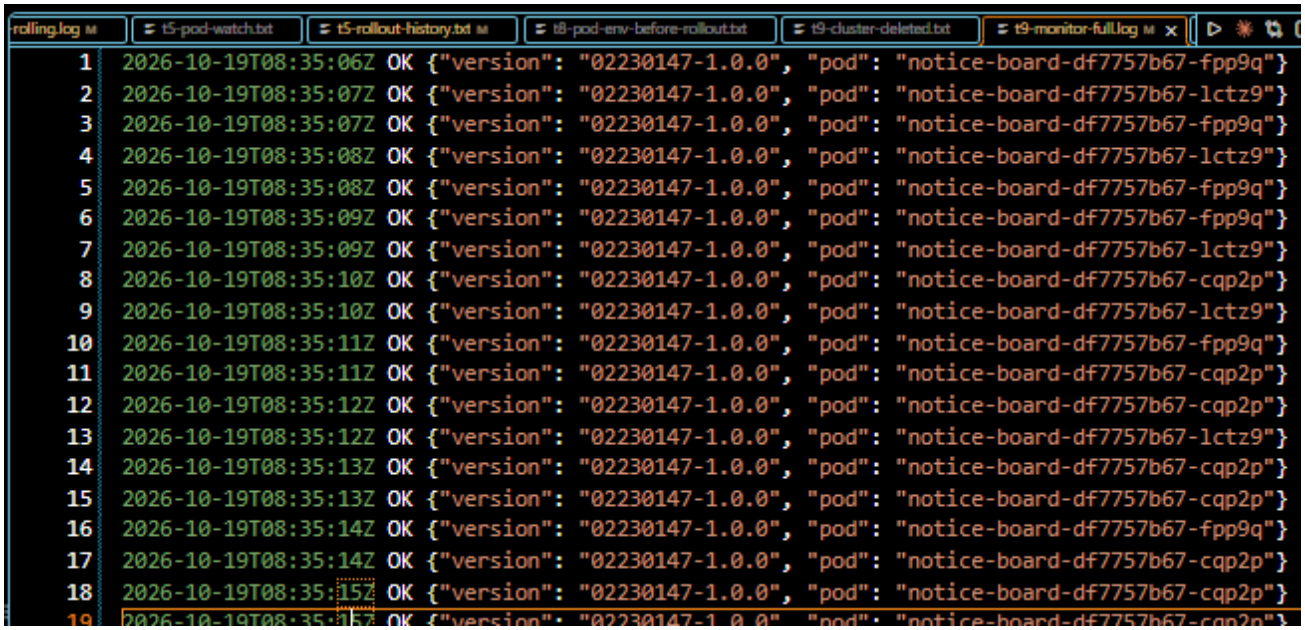
```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl rollout status deployment/notice-board-green
Waiting for deployment "notice-board-green" rollout to finish: 0 of 1 updated replicas are available...
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork>
```

```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl get pods,svc,deploy -n nb-02230147 > evidence/t9-rebuild.tx
t
```

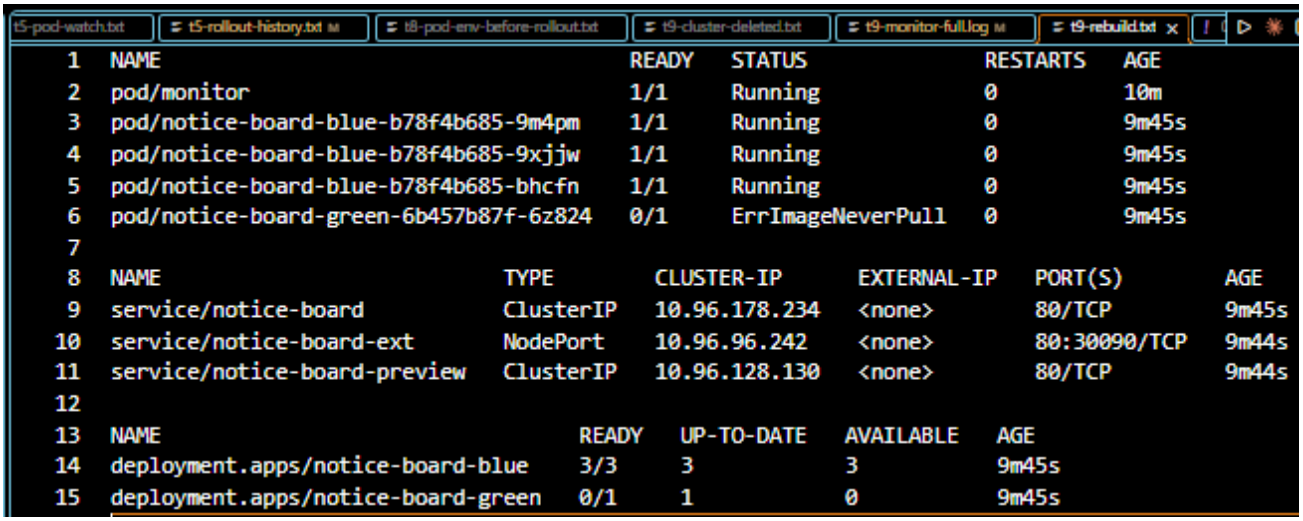
```
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kubectl config set-context --current --namespace=default
Context "kind-cte412-02230147" modified.
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind delete cluster --name cte412-02230147
Deleting cluster "cte412-02230147" ...
Deleted nodes: ["cte412-02230147-worker2" "cte412-02230147-worker" "cte412-02230147-control-plane"]
PS C:\Users\dell\OneDrive\Desktop\assignment14\kubernetes-labwork> kind get clusters | Out-File -Encoding ascii evidence/t9-cluster-de
leted.txt
```

Evidence:

```
1   class403
2   demo
3   dso202
```



```
rolling.log M | t5-pod-watch.txt | t5-rollout-history.txt M | t8-pod-env-before-rollout.txt | t9-cluster-deleted.txt | t9-monitor-full.log M
1  2026-10-19T08:35:06Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
2  2026-10-19T08:35:07Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
3  2026-10-19T08:35:07Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
4  2026-10-19T08:35:08Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
5  2026-10-19T08:35:08Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
6  2026-10-19T08:35:09Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
7  2026-10-19T08:35:09Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
8  2026-10-19T08:35:10Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
9  2026-10-19T08:35:10Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
10 2026-10-19T08:35:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
11 2026-10-19T08:35:11Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
12 2026-10-19T08:35:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
13 2026-10-19T08:35:12Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-lctz9"}
14 2026-10-19T08:35:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
15 2026-10-19T08:35:13Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
16 2026-10-19T08:35:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-fpp9q"}
17 2026-10-19T08:35:14Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
18 2026-10-19T08:35:15Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
19 2026-10-19T08:35:15Z OK {"version": "02230147-1.0.0", "pod": "notice-board-df7757b67-cqp2p"}
```



```
t5-pod-watch.txt | t5-rollout-history.txt M | t8-pod-env-before-rollout.txt | t9-cluster-deleted.txt | t9-monitor-full.log M | t9-rebuild.txt
1  NAME                                    READY   STATUS             RESTARTS   AGE
2  pod/monitor                             1/1     Running            0          10m
3  pod/notice-board-blue-b78f4b685-9m4pm   1/1     Running            0          9m45s
4  pod/notice-board-blue-b78f4b685-9xjjw   1/1     Running            0          9m45s
5  pod/notice-board-blue-b78f4b685-bhcfn   1/1     Running            0          9m45s
6  pod/notice-board-green-6b457b87f-6z824  0/1     ErrImageNeverPull  0          9m45s
7
8  NAME                          TYPE        CLUSTER-IP      EXTERNAL-IP   PORT(S)        AGE
9  service/notice-board          ClusterIP   10.96.178.234   <none>        80/TCP         9m45s
10 service/notice-board-ext      NodePort    10.96.96.242    <none>        80:30090/TCP   9m44s
11 service/notice-board-preview  ClusterIP   10.96.128.130   <none>        80/TCP         9m44s
12
13 NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
14 deployment.apps/notice-board-blue   3/3     3            3           9m45s
15 deployment.apps/notice-board-green  0/1     1            0           9m45s
```

Github URL

https://github.com/SherabDorji10/kubernetes_02230147.git